# AI Course Project Guidelines

Semester-Wide Project

**Course: Artificial Intelligence** **Project Type: Group Project**

The project will be developed progressively throughout the semester.

## Project Objective

Artificial Intelligence is evolving at an unprecedented pace. New models, tools, and applications emerge almost every day!! While understanding AI concepts is essential, the real value lies in being able to apply those concepts to build complete, intelligent systems that solve meaningful problems.

This semester-long project is designed with that philosophy in mind. Each team will be assigned a broad problem domain and will progressively transform it into a well-defined AI system by integrating the concepts learned throughout the course. Rather than implementing isolated algorithms, the goal is to understand ***when*, *why*, and *how*** different AI techniques work together in a real application.

A well-executed project can become an important part of your CV. It can be showcased during internships and placement interviews, or mentioned in higher-study applications. We therefore encourage you to treat this project as an opportunity to build something that demonstrates your understanding, creativity, and engineering skills.

This project contributes **25%** of your overall course grade (**10% Group Component + 15% Individual Component**). This means the project alone has the potential to improve your final course grade by nearly **two letter grades** or, reduce it by a similar margin. Start early and do not wait for the last one week before submission deadline!

Teams that produce high-quality, well-motivated, and technically sound work may have the opportunity to further develop their projects into research prototypes. Depending on the novelty of the idea, experimental validation, and your own interest in research, we can explore transforming such projects into publications at reputed conferences. In previous years, students who worked sincerely on research problems and internships in our lab successfully published their work at top-tier (CORE A/A*) venues. Such examples are: Multi-Agent LLM Tutoring System, or Multimodal Story Generation which were developed through dedicated UG student research effort within 2-3 months.

## Mandatory AI Components

Every project must include meaningful implementations of the following five modules:

| Module | Expected Role in the Project |
|---|---|
| **Search** | State-space formulation, exploration, optimisation, path finding, scheduling, allocation, or decision search. |
| **Planning** | Goal-oriented action sequencing, task decomposition, plan generation, or execution monitoring. |
| **Uncertainty** | Probabilistic reasoning, Bayesian modelling, uncertain observations, noisy inputs, or risk-aware decision-making. |
| **Reinforcement Learning** | Sequential decision-making through rewards, policies, environments, or adaptive behaviour. |
| **Generative AI** | Use of language models or multimodal generative models for reasoning, interaction, explanation, generation, orchestration, or decision support. |

The five modules should be connected to the assigned problem. Submitting five unrelated demonstrations or independent algorithm implementations will not be considered a complete integrated project.

## Problem Formulation and Extension

The broad application domain will be provided by the course instructor. Each team must study the domain and convert it into a precise project problem.

A strong project should clearly explain the target users, inputs, outputs, constraints, challenges, decision variables, and expected intelligent behaviour. Teams should also identify what is missing in existing approaches and propose a meaningful extension.

The extension may involve a new task formulation, an additional constraint, a new agent capability, a multimodal component, personalised behaviour, uncertainty-aware reasoning, an improved planning mechanism, or a more realistic evaluation setting.

**Research-Paper-Inspired Extensions**

Relevant research papers will be discussed and shared during the course. Students are strongly encouraged to read these papers and use them as motivation for improving their assigned projects. A project that thoughtfully adapts, extends, or experimentally investigates an idea from a recent research paper may receive **additional marks**. However, merely reproducing a paper without understanding, modification, or integration into the assigned problem will not be sufficient.
The project report and presentation must clearly mention:
*Which paper inspired the extension, what idea was adopted, what was modified, and how the proposed implementation differs from the original work.*

## Minimum Deliverables

The minimum submission must include a complete and organised code repository, a working implementation, a clear `README.md`, a final report, and a project demonstration.

The `README.md` must explain the project objective, repository structure, installation procedure, dependencies, execution commands, dataset or model requirements, and sample inputs and outputs.

The final report should document the problem formulation, system architecture, module-wise implementation, integration strategy, experiments, results, limitations, and team-member contributions.

**Mandatory Execution Requirement**

**If we cannot run the submitted project using the instructions provided in the repository, no implementation marks will be awarded.**

The code must run without requiring undocumented changes, missing files, private credentials, hard-coded local paths, or manual correction by the evaluators.

## Codebase and Reproducibility

A clean and understandable codebase is the bare minimum expectation. The repository should have a logical folder structure, modular implementation, meaningful function and variable names, and sufficient comments for non-obvious design decisions.

All dependencies must be listed explicitly. Required datasets, model checkpoints, API requirements, and configuration files must either be included or linked properly. Random seeds and evaluation settings should be documented wherever applicable.

Each module should be identifiable in the repository, and the connection between different modules should be reflected in the system architecture.

## User Interface (Optional but Recommended)

A graphical interface is not mandatory. A well-designed command-line interface is acceptable if the complete functionality can be demonstrated clearly.

However, a simple web interface, dashboard, interactive visualisation interface can significantly improve the clarity and usability of the project. A meaningful UI may receive additional appreciation when it helps demonstrate the system.

## Characteristics of a Strong Project

**What Makes a Project Strong?**

A strong project begins with a precise and relevant problem definition. It should show that the team has understood the assigned domain and has gone beyond a straightforward classroom implementation.
The system should have a coherent architecture in which search, planning, uncertainty, reinforcement learning, and generative AI each perform a justified role. Algorithm choices should be explained rather than used only because they were taught in class.
The project should include meaningful experiments, appropriate evaluation metrics, comparisons with simpler baselines, and analysis of failure cases. The final demonstration should show the complete working pipeline.
Projects that incorporate a thoughtful idea inspired by recent research literature, introduce a

meaningful new component, or investigate an interesting research question may receive additional credit.

## Projects That Will Receive Lower Marks

Projects will receive lower marks if the problem remains vague, the modules are implemented independently without integration, the code cannot be executed, the repository is incomplete, or the submitted system is mostly copied from an online source.

Marks will also be reduced if the team cannot justify its algorithmic choices, explain the code during the viva, analyse the obtained results etc.

Using AI coding tools is permitted as development assistance. However, students must understand and be able to explain every submitted component. Unverified AI-generated code, unnecessary complexity, fabricated results, or superficial "AI slop" will be heavily penalised.

## Academic Integrity and Team Responsibility

All external code, datasets, libraries, models, tutorials, and research papers must be acknowledged appropriately.

Open-source libraries may be used, but the core AI components expected under the project must be implemented and understood by the team.

Every team member should contribute meaningfully and should be able to explain the complete system during the evaluation. Individual marks will vary depending on contribution and viva performance.

**Final Note**

The goal of this project is to build something that is technically sound, practically useful, and intellectually interesting.
Do not treat the project as five disconnected assignments. Develop one coherent AI system, refine the assigned domain into a strong problem statement, study relevant research, and demonstrate how different AI techniques work together to solve the problem.

**Innovation and understanding will be rewarded.**